Lab 1

The purpose of lab 1 is to create a simulation model to study competition and firm localization in a market.

The model is based on a modified version of Hotelling's location model from 1929 (Hotelling, 1929). The model is used to analyze the behavior of firms competing in a market by choosing their geographical location relative to competitors and potential customers. A key conclusion in the model is a principle often called Hotelling's Law or the "Principle of Minimum Differentiation". The principle states that in many situations, it is rational for competing firms to strategically choose to locate close to one another in a given market. This creates a situation where both firms become equally attractive to consumers, who then choose the nearest firm to minimize their own cost or effort.

A classic example of Hotelling's model is the placement of two competing ice cream vendors on a beach. If one vendor places themselves at one end of the beach and the other at the opposite end, they will divide the beach visitors between them. However, if both place themselves in the middle of the beach, each vendor will have access to the entire customer base, as visitors will choose the nearest vendor. The principle can be used to explain why, for instance, similar stores choose to open close to each other rather than spreading out (e.g., why ICA and Coop are both located next to each other in Ålidhem). The principle does not only pertain to geographical location but also product design (e.g., multiple manufacturers producing similar products).

A problem with Hotelling's model is that the result does not hold in situations where firms compete on both price and location. One way to model this is to view firms' choices as a sequential game where they first choose a location and then set prices. As shown by d'Aspremont et al. (1979), the pricing game lacks a Nash equilibrium in pure strategies if firms are located too close to each other (d'Aspremont et al., 1979). This means that the equilibrium solution proposed in Hotelling's article *"Stability in Competition"* does not exist if firms also compete on price. Hotelling's Law therefore only applies in certain situations. Several articles have studied variations of Hotelling's model to examine in which situations the principle is valid and when it is not (Biscaia et al., 2013). Much of this literature builds on mathematical analysis of various economic models using calculus and algebra. A formal description of the model can be found in (Puu, 2002).

An alternative approach to study the model is to use numerical methods and simulation instead. An example of such an approach is this model implemented in the simulation software NetLogo (Ottino, Stonedahl, & Wilensky, 2009).

The goal of this lab is to create a simulation model in any programming language (preferably Python or Matlab) to study firm behavior in a competitive market where firms compete on both price and location. Start from the proposed model structure described below and make necessary assumptions where required. Using the model, you will investigate how different model assumptions affect the model's properties and behavior.

The lab is divided into different tasks. Each completed task gives points. To achieve a passing grade (G/3) on the lab, at least 6 points are required. The lab can give a total of 12 points.

Basic model without pricing

Start with the model in Hotelling (1929) and create a discrete simulation model of this model. You can take inspiration from the NetLogo model (https://ccl.northwestern.edu/netlogo/models/Hotelling'sLaw).

Two pizza restaurants sell pizza to local customers living along a straight road. The two firms $i = 1,2$ are located at $x_1, x_2 \in [-1,1]$. In the basic model, the firms charge the same price for pizzas, i.e., $p_1 = p_2 = p$. Along the road, n customers are evenly distributed. Each customer $j = 1, \dots, n$ has a demand function $D_j = \frac{D}{n}$ and chooses to buy from the pizzeria nearest to their location x_j. (D is total demand. A simple alternative is to set this parameter to 1. The reason demand is a function of n is to ensure total demand does not change when the granularity of the road is adjusted, which may become important in later steps when other parameters are introduced.)

Create a discrete simulation model where the road is divided into a discrete number of positions in the interval $x \in [-1,1]$. Each such position contains one customer and is a possible location for one (or more) pizzeria firms.

Calculate the profit for each firm as a function of their locations, i.e., x_1, x_2. Both firms aim to maximize their profit. Use an iterative approach to study how the firms step-by-step adjust and try to optimize their location to maximize profit, given the previous locations of the other firms. At each time step, one (or several) firms can adjust their location by moving one step towards 1, one step towards -1, or staying in place. This can, for example, be done by evaluating the firm's profit at points $x_u - \Delta x, x_i + \Delta x$, where Δx is the distance between two adjacent positions on the grid.

What happens in the model? Does the process converge to a point? How do the results from the simulation model compare to the results in Hotelling (1929)? Also compare with the NetLogo model, where two firms can only move in one dimension.

(2 points)

Basic model with pricing in a sequential game

Extend the model by allowing the pizzerias to compete by adjusting their prices. Assume that the pizzerias have a linear cost function (constant unit cost) $c(Q_i) = c_0 Q_i$. Each customer j purchases from the pizzeria i that offers the lowest total cost tc_j to buy pizza and travel to the location, i.e., $tc_{ij} = p_i + k \cdot |x_i - x_j|$ and $tc_j = \min_i tc_{ij}$. We also assume that customers have a maximum price they are willing to pay for a pizza ($D_j = \frac{D}{n}$ if $p_{i_j} \leq p^{max}$ and $D_j = 0$ if $p_{i_j} > p^{max}$) where $i_j = \operatorname{argmin}_i tc_{ij}$ is the location chosen by customer j.

Divide the price dimension into a discrete number of price steps $p_i = 0, \Delta p, 2\Delta p, \dots, p^{max}$ and let the pizzerias choose one of these prices.

First consider a sequential game where the firms first choose location and then choose prices in a subgame where the location is fixed. Analyze this game. For which location pairs do the price setting subgame have an NE with $p_i > c_0$?

Begin with a **local adjustment algorithm** with two loops. In the outer loop, each firm take turn to adjust their location. After a firm has changed their location, there is an inner loop where the firms can adjust their prices. At each time step in the inner loop, one (or several) firms are allowed to adjust their prices with a small amount to increase profit. With a local search this means to evaluate three different options $(x_i, p_i - \Delta p), (x_i, p_i), (x_i, p_i + \Delta p)$ to maximize profit where Δp is a price step in the grid.

To simplify the simulation, assume that all firms are myoptic meaning that they take the price and location of the other firms as given. In the outer loop, each firm can change their location with a small amount, i.e. choose between $(x_i - \Delta x, p_i^*), (x_i, p_i^*), (x_i + \Delta x, p_i^*)$ where Δx is a small location step and p_i^* is the NE price in the inner game. Calculate the profit for each firm as a function of both their locations $(x_1, x_2)$and the corresponding prices in the inner subgame $p_1^*(x_1, x_2)$ and $p_2^*(x_1, x_2)$. Describe the game and search for SPE in the outer location game.

(2 points)

Basic model with pricing in a simultaneous game

Assume now that the firms can adjust their location and price simultaneously. At each time step, one (or several) firms can adjust their location and prices by taking a small step in both price and location given the position (in location and price) of the other firm(s) fixed. With a local search, this means evaluating nine different options: $(x_i - \Delta x, p_i - \Delta p), (x_i - \Delta x, p_i), (x_i - \Delta x, p_i + \Delta p), \dots, (x_i + \Delta x, p_i + \Delta p)$ where Δx is a location step and Δp is a price step in the grid. What happens? Does the model converge? If so, to what point?

The firms' action choices in this version can be described as a matrix containing all combinations of locations and prices. Calculate the profit for each firm as a function of both their locations and prices, i.e., x_1, x_2, p_1, p_2.

(2 points)

Global search

Extent the model by allowing firms to take larger steps and, at each time step, move to the best location and price level given the current location and price level of the other firm. A simple but computationally intensive way to do this is to calculate the profit for all combinations of locations and prices in the action matrix at each step, assuming the other firm has a certain position in the matrix. What happens? Does the model converge? Is a different result obtained with a global adjustment algorithm compared to a local adjustment algorithm? Compare with the properties of the Hotelling model in NetLogo.

(2 points)

Elastic Demand

Let demand for each customer be a function of the total cost tc_j. With a linear demand function, demand becomes $D_j = \frac{D}{n} \cdot \left(1 - e \cdot tc_j\right)$ for $tc_j \leq \frac{1}{e}$ and $D_j = 0$ otherwise. Investigate how firms' behavior depends on different values of e. How does the model change with elastic demand compared to earlier models with static demand?

(1 points)

Plane City Instead of Linear

Extend the model by replacing the line with a plane. The model now consists of an area where customers are uniformly distributed over the interval $-1 \leq x \leq 1$ and $-1 \leq y \leq 1$. Each customer buys pizza from the pizzeria with the lowest total cost $tc_{ij} = p_i + k \cdot \sqrt{\left(x_i - x_j\right)^2 + \left(y_i - y_j\right)^2}$. Initially, there are m pizzerias located at various positions. Each pizzeria can be described by a vector (x_i, y_i, p_i) where (x_i, y_i) is the location of the pizzeria and p_i is the price for a pizza. Describe the assumptions made and justify them. What happens in the model?

(1 point)

Entry and Exit

Develop the linear model by allowing firms to enter or leave the market depending on their profitability. Modify the firms' cost function by introducing a fixed cost $c(Q_i) = c_0 Q_i + c_F$. Allow each firm to have a buffer that can be used to absorb short-term losses. If a firm runs out of buffer, it exits the market. Similarly, a new firm can attempt to enter the market if it believes it has an opportunity to make a profit. Describe and justify the assumptions made. Refer to game-theoretic articles on entry and exit where possible. What happens in the model? How many pizzerias can simultaneously fit in the market?

(1 point)

Repeated Games and Folk Theorem

Start from the description of the Folk Theorem in the course book (Chapter 16) and examine the possibility for firms to form long-term cartels in a game with two firms. First, conduct theoretical calculations to evaluate the possibility in a game with pricing, two firms, and static demand. An important concept in the theorem is to calculate minimax strategies where one firm chooses a price and location to minimize the profit of the other firm. Then develop the model to numerically calculate these values. How can the model be modified to simulate long-term considerations and possible strategies for cooperation?

(1 point)

Presentation of results:
Report the results in the form of a **PM** and a **presentation**. The task will also be presented orally at a seminar on **October 6th, 13:15-15:00** in MIT.A.356. Prepare a short presentation for the seminar.

Supervision:
September 22th, 13:15-15:00 in MIT.A.131.